AZURE EXAM REVISION GUIDE

Clear explanations • Real-world examples • Exam-focused comparisons

Topics covered

Azure Bastion • Microsoft Purview • Azure Data Box • Blob access tiers • AzCopy • Azure Files • Azure File Sync • ARM templates • Azure Arc • ExpressRoute • VPN Gateway • Container Apps • Container Instances • Availability sets • Availability zones • Logic Apps • Application Gateway

How to use this guide
Read the simple explanation first, then focus on the “Exam clue” and comparison tables. The guide explains concepts rather than memorizing portal steps. Azure features and pricing can vary by region, storage type, and SKU, so always check the current Microsoft documentation for deployment details.

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1. Azure management, security and governance

Azure Bastion

What it is: A managed service that lets administrators securely connect to Azure virtual machines using RDP (Windows) or SSH (Linux), usually through the Azure portal or a native client. The connection is carried over TLS/HTTPS (port 443).

Why use it: You do not need to give each VM a public IP address or expose RDP port 3389 / SSH port 22 directly to the internet. Bastion is deployed into your virtual network and acts as a secure jump point to VMs using their private IP addresses.

Example: A company has private Windows and Linux VMs in a VNet. An administrator opens the Azure portal, selects Connect through Bastion, and securely signs in without making the VMs publicly reachable.

Exam clue: Secure browser-based RDP/SSH to a VM **without a public IP on the VM** = Azure Bastion. Bastion is for administrator access to VMs; it is not a VPN that connects an entire office network to Azure.

Microsoft Purview

What it is: Microsoft's portfolio of data governance, data security, and compliance capabilities. It helps an organization understand what data it has, where it is stored, who can access it, and how it is protected or used.

Main ideas: Data governance helps discover, catalog, classify, and manage data. **Data security** helps protect sensitive information through capabilities such as information protection and data loss prevention. **Compliance** supports audit, records management, eDiscovery, and regulatory requirements.

Example: A bank wants to locate files containing customer identity information across its data estate, classify them as sensitive, and apply policies to reduce accidental sharing.

Exam clue: Discover/catalog/classify data, understand data lineage, protect sensitive information, or support compliance = Microsoft Purview. It is not a firewall or a backup service.

Azure Resource Manager (ARM) templates

What it is: JSON files that describe the Azure resources you want and their configuration. They let you deploy infrastructure as code instead of creating every resource manually.

Why use it: Repeatable deployments, consistent environments, version control, and automation. You can describe resources such as virtual networks, VMs, storage accounts, and their settings in a template.

Example: A development team needs the same VNet, subnet, NSG, and VM setup for development, testing, and production. A parameterized ARM template can deploy the same design with different names or sizes.

Exam clue: Declaratively define and repeatedly deploy Azure infrastructure using JSON = ARM template. Bicep is a higher-level language that compiles to ARM templates.

Azure Arc

What it is: A way to bring resources outside Azure—such as on-premises servers, other-cloud servers, and Kubernetes clusters—into Azure's management plane.

Why use it: Apply familiar Azure inventory, governance, policy, security, and monitoring practices across hybrid and multicloud environments. Arc does not physically move those servers into an Azure datacenter; it connects and represents them for management.

Example: A company has Windows servers in its own datacenter and Kubernetes clusters in another cloud. Azure Arc helps manage and govern them through Azure.

Exam clue: Manage on-premises / multicloud servers or Kubernetes using Azure tools = Azure Arc.

2. Azure storage and data movement

Azure Data Box

What it is: A physical, secure storage device that Microsoft ships to a customer to transfer very large amounts of data to or from Azure. The customer copies data onto the device, ships it back, and Microsoft uploads it to the selected Azure destination.

When to use it: When transferring terabytes or petabytes over the internet would take too long, cost too much, or be impractical because of limited network bandwidth. It is designed for offline data transfer.

Example: A media company has 80 TB of video archives at an office with a slow internet connection. It can use a Data Box device rather than upload all files over the network.

Exam clue: Very large data transfer using a shipped physical device = Azure Data Box. For normal online command-line copying, think AzCopy.

Blob Storage access tiers: Hot, Cool, Cold and Archive

These tiers help control the cost of storing blob data based on how often you expect to access it. In general, the less frequently accessed tiers have a lower storage cost but a higher cost or delay when you read or move data.

Tier	Best for	Storage cost	Access / retrieval	Minimum retention
Hot	Frequently accessed active data	Highest of the online tiers	Lowest access/transaction cost; millisecond access	No minimum
Cool	Infrequently accessed data that still needs immediate access	Lower than Hot	Higher access and retrieval charges than Hot; millisecond access	30 days
Cold	Rarely accessed data that still needs immediate access	Lower than Cool	Higher access and retrieval charges than Cool; millisecond access	90 days
Archive	Rarely accessed long-term data, such as old records	Lowest storage cost	Offline; must be rehydrated before reading, typically hours; highest retrieval/operation costs	180 days

Important cost rule: Hot = pay more to store, less to access. Cool/Cold = cheaper to store, more expensive to access. Archive = cheapest to store, but offline and expensive/slower to retrieve. Actual prices depend on region, redundancy, account type, capacity, transactions, retrieval, and early-deletion charges.

Example: Current website images belong in Hot. Last year's reports that may be opened occasionally fit Cool. Old project files that are rarely needed but must be available quickly fit Cold. Legal records that are almost never opened and can wait hours fit Archive.

Exam trap: Archive is not simply “the slowest online tier”—it is offline. You must rehydrate it to an online tier before reading. Cool and Cold are online tiers and allow immediate access.

Storage redundancy and the 16-nines question

Redundancy means Azure keeps multiple copies of data to protect against hardware, datacenter, zone, or regional failures. It is about durability and availability—not a substitute for backup against accidental deletion or malicious changes.

Option	Where copies are kept	Durability	Key point
LRS	One datacenter in primary region	11 nines	Lowest cost; limited datacenter-disaster protection
ZRS	Across separate availability zones in primary region	12 nines	Protects against a zone/datacenter outage in the region
GRS / RA-GRS	Primary region plus secondary region	16 nines	Geo-replication; RA-GRS allows reads from secondary
GZRS / RA-GZRS	Across zones in primary region, plus secondary region	16 nines	Combines zone and regional protection; RA-GZRS adds secondary read access

Answer to your multiple-choice question: Both GZRS and RA-GZRS provide 16-nines durability and combine zone protection in the primary region with replication to a secondary region. If the question asks which option provides the strongest combination of zone-level and regional disaster protection, choose **Geo-zone-redundant storage (GZRS)**. If it specifically requires read access to the secondary region, choose **RA-GZRS**. GRS and RA-GRS also have 16-nines durability, but do not provide zone-redundant replication in the primary region.

AzCopy

What it is: A command-line tool for copying data to, from, or between Azure Storage accounts. It supports transfers involving Blob Storage and Azure Files.

Why use it: Efficiently transfer many files, automate repeatable transfers with scripts, and move data between local machines and Azure storage. Authentication can use Microsoft Entra ID or a suitable SAS token, depending on the scenario.

Example: An administrator uses AzCopy to upload a local folder to a blob container, or copy blobs from one storage account to another.

Exam clue: Command-line utility to copy data between local storage and Azure Storage = AzCopy.

Azure Files

What it is: A fully managed cloud file share. It provides familiar shared folders using standard SMB (commonly Windows) and NFS (commonly Linux) protocols, depending on the share type and configuration.

Why use it: Multiple VMs, users, or applications can access the same shared files. It can replace or extend a traditional file server without managing the storage hardware.

Example: Several Azure VMs need a common shared folder for application files or team documents. Mount an Azure file share rather than maintaining a separate file server.

Exam clue: Managed cloud file share accessible using SMB/NFS = Azure Files.

Azure File Sync

What it is: A service that synchronizes Windows Server file shares with a centralized Azure file share. On-premises servers can keep local copies while Azure Files acts as the central cloud location.

Cloud tiering: Frequently used files remain cached locally. Less frequently used files can be tiered to Azure, while their names and folder structure remain visible on the server. When a user opens a tiered file, the content is recalled from the cloud.

Example: A company has branch offices with Windows file servers. File Sync keeps their data synchronized through Azure Files, avoiding the need for direct server-to-server synchronization.

Exam clue: Synchronize on-premises Windows file servers with Azure Files and optionally cache hot files locally = Azure File Sync.

Azure Files vs Azure File Sync vs AzCopy

Service/tool	Main purpose
Azure Files	Provides the cloud file share itself.
Azure File Sync	Keeps Windows Server file shares synchronized with Azure Files; supports cloud tiering.
AzCopy	Copies data using command-line commands; it is not a continuously synchronizing file-server service.

3. Hybrid connectivity and compute

ExpressRoute

What it is: A private, dedicated connectivity option that extends an organization's on-premises network to Microsoft cloud services through a connectivity provider. Traffic does not travel over the ordinary public internet path.

Why use it: For predictable performance, higher bandwidth options, consistent latency, and private connectivity for important enterprise workloads. ExpressRoute circuits have redundant connections at the Microsoft edge, but customers should design end-to-end resiliency too.

Example: A large organization needs a reliable private network connection from its datacenter to Azure for business-critical systems.

Exam clue: Private dedicated connectivity from on-premises to Azure that bypasses the public internet = ExpressRoute.

VPN Gateway

What it is: A virtual network gateway that creates encrypted VPN connections between Azure and another network over the public internet, or between Azure VNets.

Common connection types: Site-to-site (S2S) connects an entire on-premises network to an Azure VNet using an IPsec/IKE tunnel. **Point-to-site (P2S)** lets an individual remote computer connect to the VNet. **VNet-to-VNet** connects Azure VNets through VPN tunnels.

Example: Employees working from home connect their laptops to the company's Azure VNet using P2S VPN. A branch office can use S2S VPN.

Exam clue: Encrypted VPN tunnel over the internet = VPN Gateway. Private provider connection not over the public internet = ExpressRoute.

Feature	ExpressRoute	VPN Gateway
Path	Private connection via connectivity provider	Encrypted tunnel over public internet (typical use)
Encryption	Private does not automatically mean encrypted; additional encryption can be configured where required	IPsec/IKE encryption for common S2S connections
Typical fit	Enterprise-critical, predictable connectivity	Branch offices, remote users, VNet-to-VNet connections
Cost / setup	Usually higher cost and provider coordination	Often simpler and lower initial cost

Useful extra: A site-to-site VPN can be configured as a backup path alongside ExpressRoute. ExpressRoute and VPN Gateway solve network connectivity; Azure Bastion solves secure administrator access to individual VMs.

Azure Container Apps (ACA)

What it is: A managed, serverless platform for running containerized applications without managing the underlying server infrastructure or Kubernetes control plane.

What it is good for: APIs, microservices, background jobs, and event-driven applications. It supports ingress, revisions, traffic splitting, built-in service discovery, and autoscaling using HTTP traffic, events, CPU/memory, or supported KEDA scalers. Many workloads can scale to zero, depending on the scaling configuration.

Example: A web API runs in a container and automatically adds replicas when requests increase, then scales down when demand falls.

Exam clue: Managed container application platform with autoscaling, microservices, ingress, and less infrastructure management = Azure Container Apps.

Azure Container Instances (ACI)

What it is: A service for running containers directly in Azure without provisioning or managing virtual machines. It is designed for fast, straightforward container execution.

What it is good for: A quick one-off task, a short-lived process, a simple container workload, or a container that needs to start quickly. ACI does not provide the same application platform and built-in orchestration features as Container Apps.

Example: A scheduled script needs to run inside a container for a few minutes, complete its work, and stop.

Exam clue: Run a container quickly without managing VMs or a full orchestration platform = ACI.

Question	Container Apps (ACA)	Container Instances (ACI)
Main focus	Application platform for containerized apps	Run container instances / groups
Scaling	Built-in event/HTTP-based autoscaling; can scale to zero in supported scenarios	No equivalent app-level autoscaling platform built in
Best fit	APIs, microservices, jobs, event-driven apps	Simple, fast-starting container workloads and tasks
Management	More app platform features are managed for you	You manage the container group's lifecycle and configuration

4. Availability and resilience

Availability Zones

What they are: Physically separate groups of datacenters within one Azure region. Each zone has independent power, cooling, and networking. A supported region commonly has three zones, although support and service behavior vary by region and service.

Why use them: Deploy or replicate application components across zones so that a failure in one datacenter/zone does not take down the entire application. Zone redundancy protects against a zone-level failure, not necessarily a full regional outage.

Example: Deploy multiple application instances across different availability zones. If one zone loses power, instances in the other zones can continue serving users.

Exam clue: Separate datacenters inside the same region with independent power, cooling, and networking = Availability Zones.

Availability Sets

What they are: A logical grouping of Azure VMs that Azure distributes across fault domains and update domains to reduce the chance that all VMs are affected by the same local hardware failure or planned maintenance.

Fault domain (FD): A group of VMs that share a common power source and network switch. Separating VMs across fault domains reduces the risk that one hardware/power/network failure affects all of them.

Update domain (UD): A group of VMs that can be restarted together during planned platform maintenance. Azure updates one update domain at a time, allowing VMs in other update domains to keep running.

Example: Put two or more VMs in one availability set. Azure places them across fault and update domains so a single rack failure or one maintenance wave is less likely to stop both at once.

Exam clue: Fault domains + update domains = Availability Set. Separate physical datacenters = Availability Zones.

Feature	Availability Set	Availability Zones
Placement	Different fault/update domains within a datacenter/region infrastructure	Physically separate datacenter zones in one region

Feature	Availability Set	Availability Zones
Protects against	Rack, power, network hardware failures and planned maintenance grouping	A datacenter/zone outage
Key terms	Fault domains and update domains	Zone 1, Zone 2, Zone 3 (where supported)
Resilience	Lower than zones; may share datacenter-level risks	Stronger isolation; recommended for supported production workloads

Remember: Availability is about keeping a service running. Durability is about avoiding data loss. Zones help with datacenter failures; geo-redundancy helps with regional disasters. A backup helps recover deleted, corrupted, or changed data—replication alone may copy those changes too.

5. Application integration and traffic management

Azure Logic Apps

What it is: A cloud service for building automated workflows that connect apps, services, data, and systems. Workflows are often created visually with triggers, actions, conditions, and connectors, requiring little or no code.

Trigger: The event that starts a workflow (for example, a new email arrives or a file is uploaded). **Action:** A task performed after the trigger (for example, save an attachment, send a notification, or update a database).

Example: When a customer submits an online form, a Logic App can save the details, send a confirmation email, notify a staff member, and create a task in another system.

Exam clue: Automate a business process across multiple services using triggers, actions, and connectors = Logic Apps. It is workflow integration, not primarily a VM hosting service.

Azure Application Gateway

What it is: A web traffic load balancer that operates at Layer 7 (the application layer). It understands HTTP/HTTPS requests and can route traffic based on information such as the requested host name or URL path.

Key capabilities: Distributes web requests to backend servers; supports URL path-based routing (for example, /images to one backend and /api to another); can terminate SSL/TLS; supports autoscaling and zone redundancy in supported configurations; and can integrate with Web Application Firewall (WAF) to help protect web applications against common web attacks.

Example: A company hosts a website and an API. Requests to example.com/shop go to the shop backend, while example.com/api goes to the API backend. WAF can inspect requests for malicious web patterns.

Exam clue: Layer 7 HTTP/HTTPS routing, URL path-based routing, TLS termination, or WAF for a web application = Application Gateway. Azure Load Balancer is mainly Layer 4 (TCP/UDP).

6. Quick exam revision: choose the right service

Exam scenario clue	Best match
Secure RDP/SSH to private VMs without public IPs	**Azure Bastion**
Catalog, govern, classify, protect, and manage data compliance	**Microsoft Purview**
Ship a device to transfer huge data volumes offline	**Azure Data Box**
Copy files to/from/between Azure Storage using CLI	**AzCopy**
Provide a managed SMB/NFS cloud file share	**Azure Files**
Sync Windows file servers with Azure Files; cache hot files	**Azure File Sync**
Deploy repeatable Azure infrastructure from JSON	**ARM templates**
Manage on-premises and multicloud servers/Kubernetes through Azure	**Azure Arc**

Exam scenario clue	Best match
Private provider connection from datacenter to Microsoft cloud	**ExpressRoute**
Encrypted VPN over internet to Azure	**VPN Gateway**
Managed platform for APIs, microservices, autoscaling containers	**Azure Container Apps**
Run a container quickly without managing VMs	**Azure Container Instances**
VM high availability using fault and update domains	**Availability Set**
Resilience across physically separate datacenters in one region	**Availability Zones**
Automate workflows between services using triggers/actions	**Azure Logic Apps**
Layer 7 web routing, URL paths, TLS, WAF	**Application Gateway**
Lowest blob storage price; offline retrieval	**Archive tier**
16-nines durability with zones + secondary region	**GZRS / RA-GZRS**

7. Short self-test

1. You need to connect to a private Azure VM from the portal without assigning a public IP to the VM. Which service?
Answer: Azure Bastion.

2. You need to move 50 TB to Azure but the internet connection is too slow. Which service?
Answer: Azure Data Box.

3. You need to keep an on-premises Windows file server synchronized with a cloud file share and cache frequently used files locally. Which service?
Answer: Azure File Sync with Azure Files.

4. You need a private connection from a company datacenter to Azure that does not use the public internet. Which service?
Answer: ExpressRoute.

5. You need an encrypted connection from a remote employee's laptop to an Azure VNet over the internet. Which service and connection type?
Answer: VPN Gateway with Point-to-site (P2S).

6. You need to route /images and /api to different web backends and protect the web app with WAF. Which service?
Answer: Application Gateway (WAF_v2).

7. Which redundancy option combines zone replication in the primary region and replication to a secondary region?
Answer: GZRS. RA-GZRS adds read access to the secondary region.

8. You need to run a small one-off container task quickly. Which service is a strong fit?
Answer: Azure Container Instances.

9. You need an event-driven containerized API that scales based on HTTP traffic. Which service?
Answer: Azure Container Apps.

10. Which availability-set feature separates VMs for planned maintenance?
Answer: Update domains. Fault domains separate VMs by shared hardware/power/network failure boundaries.

8. Official Microsoft Learn references

Use these official pages to verify details, check regional availability, and review the latest feature limitations and pricing.

- **Azure Storage redundancy:** https://learn.microsoft.com/en-us/azure/storage/common/storage-redundancy

- **Blob access tiers:** https://learn.microsoft.com/en-us/azure/storage/blobs/access-tiers-overview
- **Azure Bastion:** https://learn.microsoft.com/en-us/azure/bastion/bastion-overview
- **Microsoft Purview:** https://learn.microsoft.com/en-us/purview/
- **Azure Data Box:** https://learn.microsoft.com/en-us/azure/databox/data-box-overview
- **AzCopy:** https://learn.microsoft.com/en-us/azure/storage/common/storage-use-azcopy
- **Azure Files:** https://learn.microsoft.com/en-us/azure/storage/files/storage-files-introduction
- **Azure File Sync:** https://learn.microsoft.com/en-us/azure/storage/file-sync/file-sync-introduction
- **ARM templates:** https://learn.microsoft.com/en-us/azure/azure-resource-manager/templates/
- **Azure Arc:** https://learn.microsoft.com/en-us/azure/azure-arc/overview/
- **ExpressRoute:** https://learn.microsoft.com/en-us/azure/expressroute/
- **VPN Gateway:** https://learn.microsoft.com/en-us/azure/vpn-gateway/vpn-gateway-about-vpngateways
- **Azure Container Apps:** https://learn.microsoft.com/en-us/azure/container-apps/overview
- **Azure Container Instances:** https://learn.microsoft.com/en-us/azure/container-instances/container-instances-overview
- **Availability sets:** https://learn.microsoft.com/en-us/azure/virtual-machines/availability-set-overview
- **Availability zones:** https://learn.microsoft.com/en-us/azure/reliability/availability-zones-overview
- **Azure Logic Apps:** https://learn.microsoft.com/en-us/azure/logic-apps/logic-apps-overview
- **Application Gateway:** https://learn.microsoft.com/en-us/azure/application-gateway/overview